OBD-II Diagnostic on STM32F429

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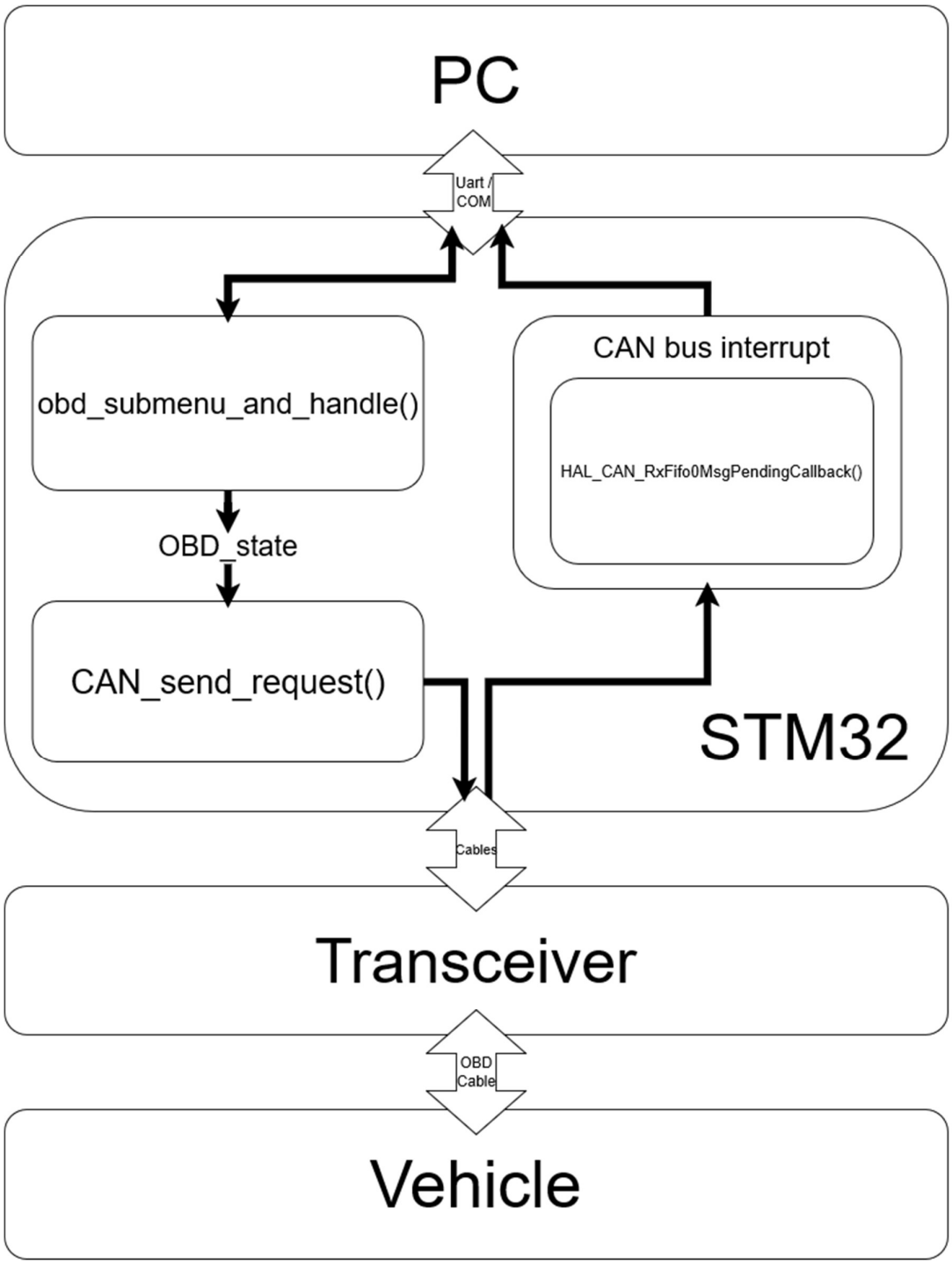
1. Hardware

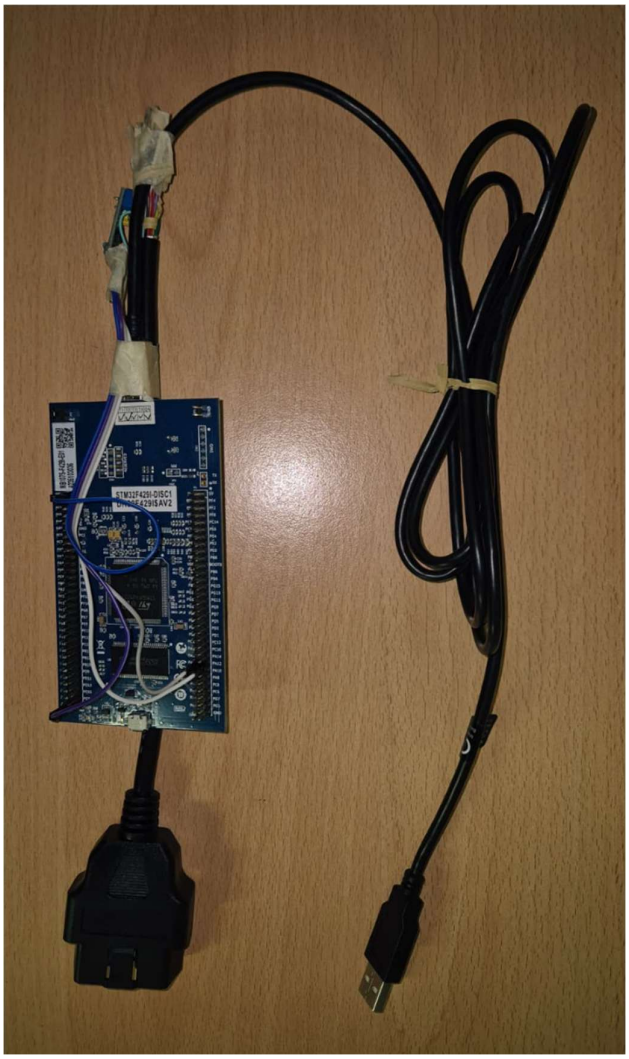
1.1 System Overview

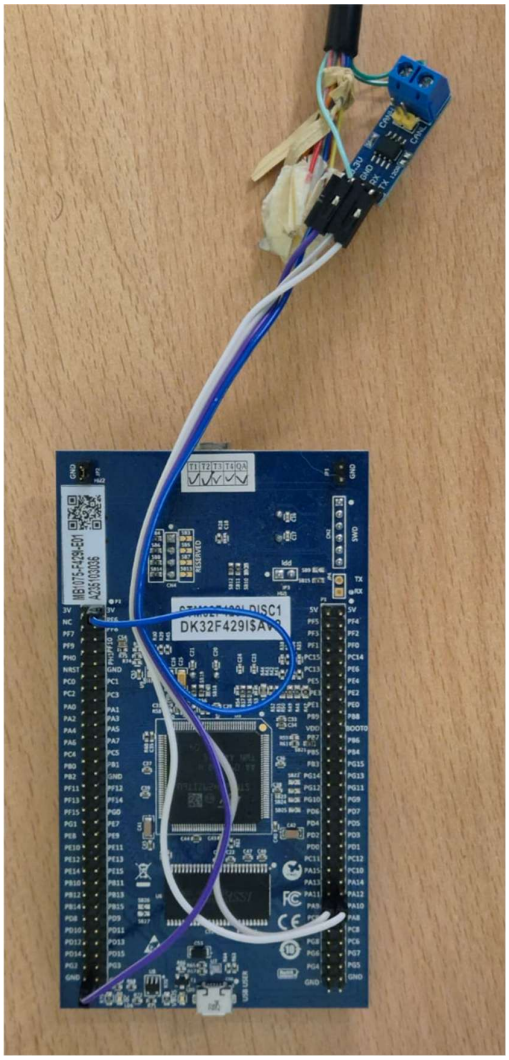
The system connects to the vehicle On-Board Diagnostic II (OBD-II) port via Controller Area Network (CAN) bus and communicates with a PC through Universal Asynchronous Receiver-Transmitter (UART).



Connections:

- OBD cable → vehicle OBD-II port
- CAN transceiver between STM32 and OBD connector
- STM32 board → PC via USB





1.2 MCU

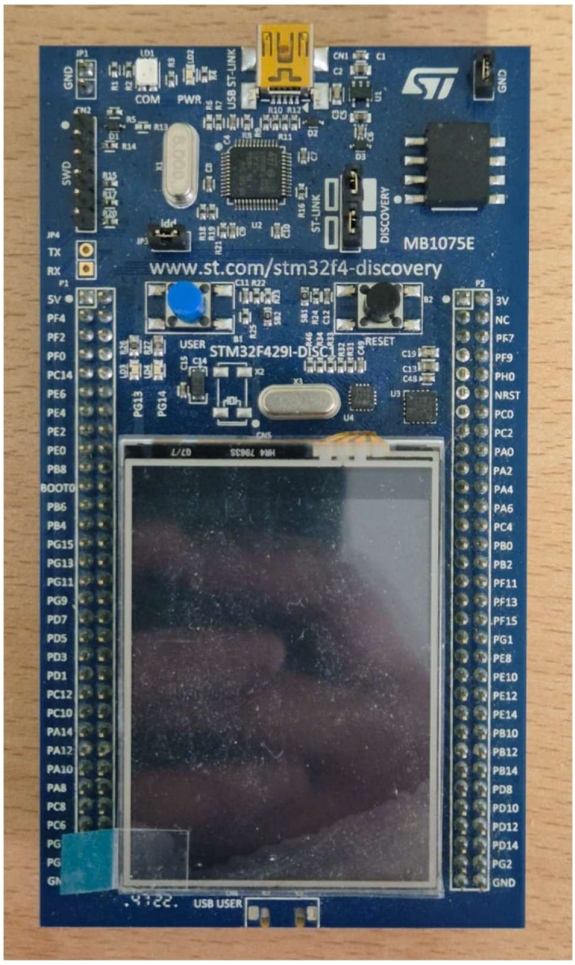
STM32 is a 32-bit microcontroller family made by STMicroelectronics.

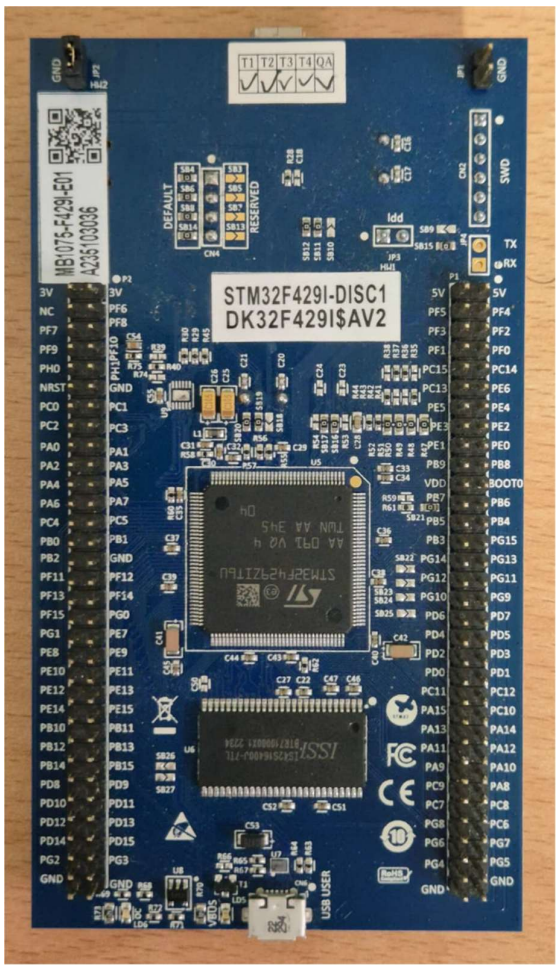
Board: STM32F429I-DISC1

MCU: STM32F429

Selected because it provides:

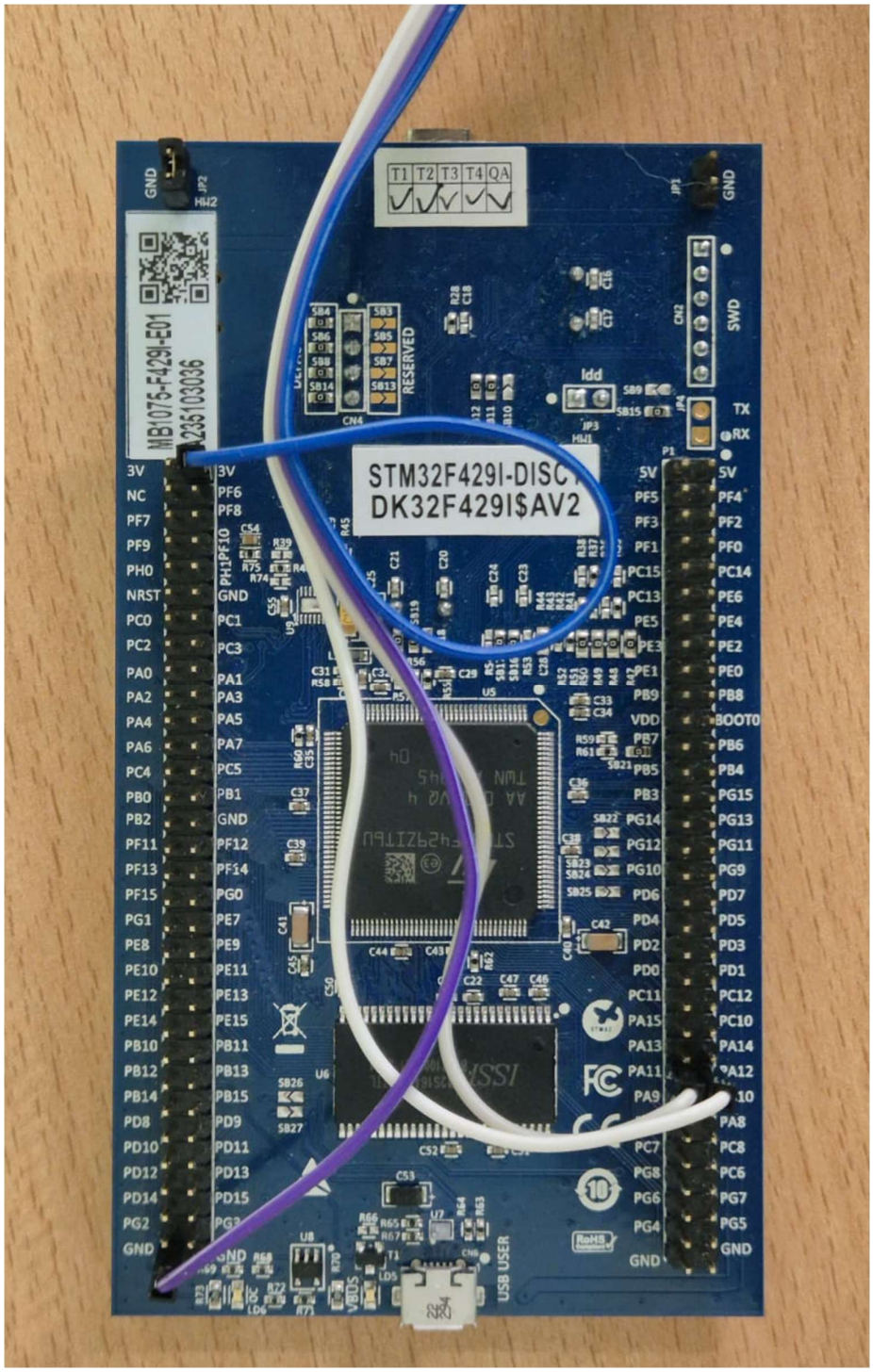
- Integrated CAN bus controller
- Universal Serial Bus (USB) interface
- Integrated ST-Link programmer
- Touchscreen (for future improvements)
- Good expansion capabilities





CAN bus pins used:

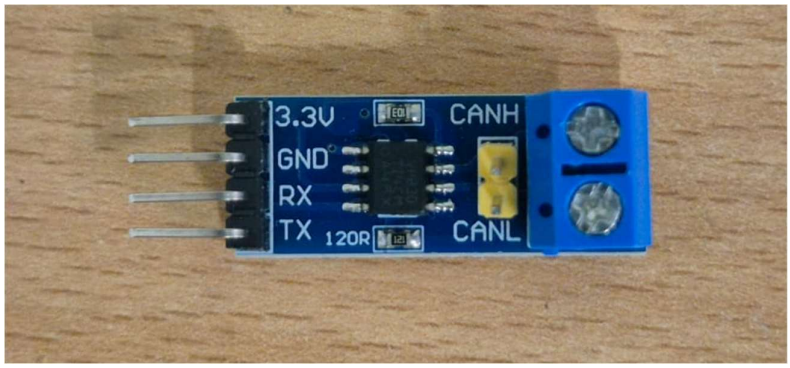
- CAN1_RX → PA11
- CAN1_TX → PA12

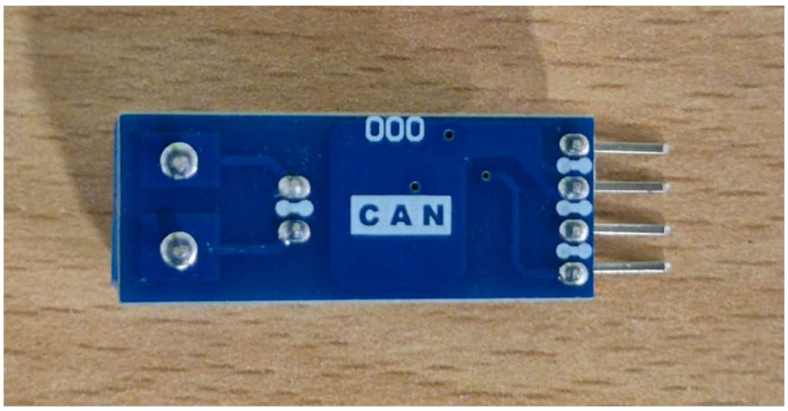


1.3 CAN Bus Transciver

Model: Texas Instruments SN65HVD230 (VP230 variant), cheap module sourced from AliExpress.

- Operates at 3.3V (compatible with STM32)
- On board 120Ω termination resistor





Connections STM32 side:

- PA11 → CAN_RX
- PA12 → CAN_TX

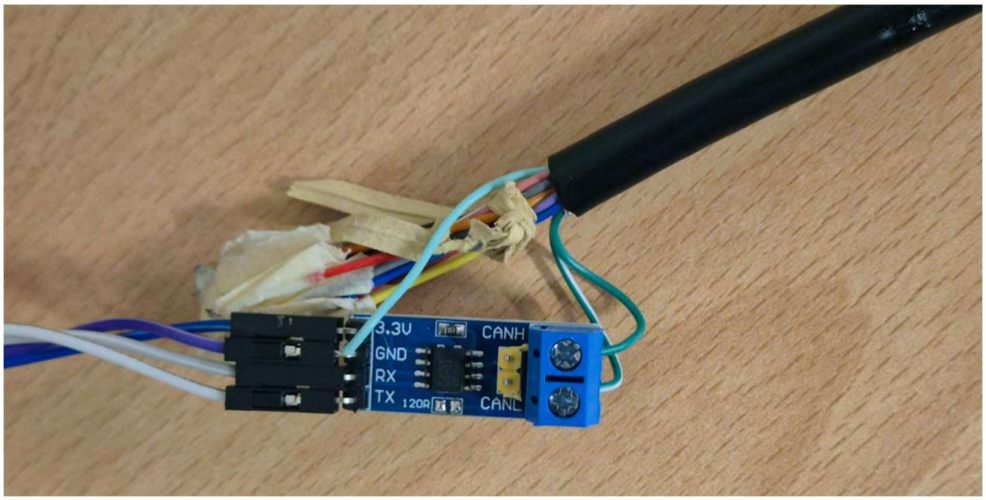
Connections OBD side:

- CANH → pin 6
- CANL → pin 14

A common ground (GND) reference was established by connecting:

- STM32 board GND
- CAN transceiver GND
- OBD-II connector Pin 5 (Vehicle Signal Ground)

This ensures proper signal reference between the vehicle CAN bus and the microcontroller system, which is required for reliable CAN communication.



1.4 OBD-II cable

Generic OBD-II cable from AliExpress, cut and stripped.

Used pins:

- Pin 6 → Vehicle CAN High
- Pin 14 → Vehicle CAN Low
- Pin 5 → Vehicle Signal Ground

1.5 Power supply

- STM32 board powered via USB (5V from PC)
- SN65HVD230 powered from STM32 3.3V rail

1.6 UART Interface

UART via integrated ST-Link Virtual Comunication (COM) Port.

Settings:

- Baudrate: 115200
- 8N1
- No flow control

Used to:

- Send commands to STM32
- Receive parsed OBD-II data
- Receive raw CAN bus frames

PC software used:

- TeraTerm
- PuTTY

2. Software

2.1 Main Loop Behavior

Inside *while(1)*:

- The system polls the selected OBD Parameter IDentification number (PID) every 500 ms via the *CAN_Send_Request()* function
- CAN bus request is transmitted
- Response is:
    - Parsed (if predefined PID)
    - Or transmitted raw over UART (generic PID mode)

To change PID selection, hardware reset button is used.

2.2 Functions

This section describes the core functions used to implement OBD-II communication over CAN bus.

2.2.1 obd_submenu_and_handle (void)

This function is called before entering the main loop. It handles user interaction via UART and selects the active OBD PID to be polled.

Description:

- Displays a submenu over UART.
- Waits for user input (single character).
- Sets a global state variable (obd_state) according to selection.
- If option "5" is selected, waits for a hexadecimal PID string (00–FF).
- Stores generic PID in generic_pid.

Behavior:

- Blocking UART reception
- Input terminates on '\r' or '\n'
- Performs minimal input validation

The menu structure is modular and can be easily extended to support additional PIDs or diagnostic services by adding new selection cases.

2.2.2 CAN_Send_Request(void)

Sends an OBD-II Mode 01 request frame over CAN bus.

Description:

- Uses standard ID 0x7DF (functional broadcast).
- Constructs an 8-byte frame:

[02] [01] [PID] [00] [00] [00] [00] [00]

- Adds message to CAN TX mailbox using:

HAL_CAN_AddTxMessage()

Notes:

- The function is called every 500 ms inside the main loop.
- PID depends on the selected state (obd_state).
- There is a similar function called "*CAN_Send_RPM_Request*" dedicated to Revolutions Per Minute (RPM) request only, used during early debugging phases to validate CAN bus communication.

2.2.3 HAL_CAN_RxFifo0MsgPendingCallback(CAN_HandleTypeDef *hcan)

Handles incoming CAN bus frames (interrupt-driven reception).

This function is not user-defined from scratch. It is a weak callback function already declared inside the STM32 Hardware Abstraction Layer (HAL) CAN driver (*stm32f4xx_hal_can.c*).

The HAL provides:

*__weak void HAL_CAN_RxFifo0MsgPendingCallback(CAN_HandleTypeDef *hcan)*

The user overrides this weak definition by implementing the same function inside the application code.

Automatically triggered when:

- A new CAN bus message is received
- FIFO0 contains at least one pending message
- Interrupt *CAN_IT_RX_FIFO0_MSG_PENDING* is enabled

The function is invoked by the HAL interrupt handler *HAL_CAN_IRQHandler()*.

The callback:

1. Reads the received CAN frame using *HAL_CAN_GetRxMessage()*
2. Checks that the response is within the Engine Control Unit (ECU) response range (0x7E8 – 0x7EF)
3. Uses a global state variable (*obd_state*)
4. Selects the appropriate parsing routine through a switch statement

This state-based design makes the parsing logic modular and easily extendable. New PIDs can be added by simply:

- Extending the *obd_state* enumeration
- Adding a new case inside the switch block

The approach is conceptually similar to the modular structure used in *obd_submenu_and_handle()*, ensuring consistency in software architecture.

3. CAN Bus Configuration

3.1 Bitrate

- 500 kbps (some older car may use 250kbps)
- CAN clock: 42 MHz

Configuration:

- Prescaler = 6
- BS1 = 11
- BS2 = 2

3.2 Filter configuration

All frames are accepted.

Inside MX_CAN1_Init function:

```
CAN_FilterTypeDef filter = {0};

filter.FilterBank = 0;

filter.SlaveStartFilterBank = 14;

filter.FilterMode = CAN_FILTERMODE_IDMASK;

filter.FilterScale = CAN_FILTERSCALE_32BIT;

filter.FilterIdHigh = 0x0000;

filter.FilterIdLow = 0x0000;

filter.FilterMaskIdHigh = 0x0000;

filter.FilterMaskIdLow = 0x0000;

filter.FilterFIFOAssignment = CAN_RX_FIFO0;

filter.FilterActivation = ENABLE;


HAL_CAN_ConfigFilter(&hcan1, &filter);

HAL_CAN_Start(&hcan1);
```

This configuration allows to receive all ECU responses (0x7E8–0x7EF)

4. OBD-II communication

4.1 Request frame

Standard OBD-II request:

- ID: 0x7DF (functional broadcast)
- Data Length Code (DLC): 8 bytes

Example:

[02] [01] [PID] [00] [00] [00] [00] [00]

Where:

- 0x02 = number of additional bytes
- 0x01 = Mode 01 (Show current data)
- PID = requested parameter

4.2 Response handling

Reception handled by interrupt callback:

*void HAL_CAN_RxFifo0MsgPendingCallback(CAN_HandleTypeDef *hcan)*

The callback:

- Reads CAN bus frame
- Verifies response ID (0x7E8–0x7EF)
- Checks Mode (0x41)
- Parses data if PID matches selected one
- Sends result via UART

5. UART menu interface

Function:

void obd_submenu_and_handle(void);

Menu options:

1 -> RPM (0x0C)

2 -> Engine Load (0x04)

3 -> Speed (0x0D)

4 -> Oil Temp (0x5C)

5 -> Inserisci PID esadecimale

Behavior:

- Single character selects predefined PID
- Option 5 allows full hexadecimal string input
- Input terminates on:
  - '\r'
  - '\n'

6. Implemented PIDs

Parsed PIDs:

- RPM (0x0C)
- Engine Load (0x04)
- Vehicle Speed (0x0D)
- Oil Temperature (0x5C)

Raw Tested PIDs:

- Coolant Temperature (0x05)
- Throttle Position (0x11)
- Intake Air Temperature (0x0F)
- Vehicle Speed (0x0D)
- Oil Temperature (0x5C)

7. Vehicles tested

- Subaru BRZ Gen 2 ZD8
- Mini R56 LCI

Note:
Oil temperature on Mini may also be available on proprietary (non-standard) PIDs.

8. Glossary

Acronym	Meaning	Page
CAN	Controller Area Network	2
COM	Communication port	7
DLC	Data Length Code	12
ECU	Engine Control Unit	9
GND	Ground	6
HAL	Hardware Abstraction Layer	9
OBD-II	On-Board Diagnostic II	2
PID	Parameter Identification number	8
RPM	Revolution Per Minute	9
STM32	32-bit microcontroller family made by STMicroelectronics	4
UART	Universal Asynchronous Receiver-Transmitter	2
USB	Universal Serial Bus	3